

RBSE 2020

Grade 12

Chemistry

Time: 3 Hours 15mins

Max. Marks: 56

SECTION – A

1 × 13 = 13

1. Write the value of packing efficiency in body-centred cubic structure.
2. Write definition of Osmosis.
3. Write the chemical equation of reaction taking place on the anode in cadmium-nickel storage cell.
4. Write Faraday's first law of electrolysis. A
5. Draw a labelled diagram of electrical dispersion method for preparation of colloidal solution.
6. Write oxidation state of Ni in $[\text{Ni}(\text{CO})_4]$
7. Write definition of ambidentate ligand.
8. Write chemical equation of Wurtz reaction.
9. Write IUPAC name of succinic acid.
10. Write chemical formula of ethyl isothiocyanate.
11. Write the name of amino acid which is optically inactive.
12. Write monomer units of polymer Dacron
13. Write definition of optical isomerism

SECTION-B

14. (a) Write any two differences between crystalline and amorphous solids.

(b) Draw the diagram showing impurity defect.

1+1=2

15. Calculate the osmotic pressure of 0.01 M solution at 27°C temperature.

($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

2

16. If the value of molar conductivities at infinite dilution for CH_3COOH , HCl and NaCl are 390.5, 425.4 and $126.4 \text{ S cm}^2 \text{ mol}^{-1}$ respectively at 298 K. Calculate the molar conductivity at infinite dilution of CH_3COONa .

2

17. (a) Draw a labelled diagram of froth floatation method.

(b) Write any two differences between calcination and roasting.

1+1=2

18. (a) Why transition elements form interstitial compounds?

(b) The atomic radii of lanthanide elements decrease on moving from left to right. Explain.

1+1=2

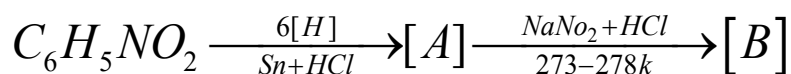
19. (a) Write IUPAC name of complex $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$

(b) Write the geometry and magnetic nature of the complex ion $[\text{NiCl}_4]^{2-}$, on the basis of valence bond theory.

1+1=2

20. (a) Alkyl amines are more basic than ammonia. Explain

(b) formula of [A] and [B] in the following sequence of reaction:



1+1=2

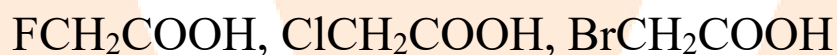
21. (a) Arrange the following haloamide in increasing order of their acidity



- (b) Why carboxylic acids are more acidic than phenol?

OR

- (a) Arrange the following haloamide in increasing order of their acidity



- (b) Aldehydes are more reactive than ketones towards nucleophilic addition reaction. Explain.

1+1=2

22. (a) Write any two differences between DNA and RNA

- (b) What happens when glucose react with bromine water?

Give chemical equation

1+1=2

23. (a) Write any two differences between condensation and addition polymerization.

- (b) Differences between biodegradable and non-biodegradable polymers

1+1=2

24. (a) Draw the structure of geometrical isomers of but-2-ene.

- (b) Draw the structure of conformer of ethane by

Newman's projection formula

1+1=2

SECTION-C

25. Read the given paragraph and write answers of the following questions:

On the basis of particle size, solutions may be divided into three types — true solutions, colloids, and suspensions. Colloid is a heterogeneous system in which one substance is dispersed (dispersed phase) as very fine particles in another substance called dispersion medium. Colloids are further classified on the basis of various criteria. Colloidal solutions exhibited various properties like — Tyndall effect, Brownian movement, electrophoresis and coagulation etc.

- (a) Write any two differences between lyophilic and lyophobic colloids.
- (b) Define the electrophoresis.
- (c) What happens when a beam of light is passed through a colloidal solution?

$$1 + 1 + 1 = 3$$

26. Read the given paragraph and write answers of the following questions:

Soaps are sodium or potassium salts of long chain fatty acids. Soaps containing sodium salts are formed by heating fat with aqueous sodium hydroxide solution. This reaction is known as saponification.

Synthetic detergents are cleansing agents which have all properties of soap, but which actually do not contain any soap. Therefore, these are also called 'soapless soap'.

Detergents are derivatives of long chain hydrocarbons and sulphuric acid. Detergents are mainly classified into three categories:

- (i) anionic
- (ii) cationic
- (iii) non-ionic detergents
- (a) Why do soaps not work in hard water?
- (b) Write any one example of cationic detergent.
- (c) Write chemical equation of saponification reaction.

1 + 1 + 1

27. (a) Write hybridized state of carbon bonded to halo gen atom in benzyl chloride.
- (b) Explain the mechanism of bimolecular nucleophilic substitution reaction.

OR

- (a) Write hybridized state of carbon bonded to halogen atom in allyl chloride.
- (b) Explain the mechanism of uni-molecular nucleophilic substitution reaction.

1+2

28. (a) Write definition of half-life for any chemical reaction.
- (b) Write any two differences between order of reaction and molecularity.
- (c) For the reaction $A \rightarrow B$, the concentration of a reactant changes from 0.05 M to 0.03 M in 20 minutes. Calculate the average rate of reaction using units of time both in minutes and seconds.

1+1+2=4

OR

- (a) Write definition of first order reaction.
- (b) Differentiate between rate of reaction and specific reaction rate.
- (c) The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if value of rate constant is $1.5 \times 10^{-4} \text{ mol L}^{-1} \text{ S}^{-1}$

29. (a) Write general electronic configuration of elements of group- 15.

(b) Why is H_2O a liquid and H_2S a gas?

(c) Complete the following reaction?



(d) Draw the structure of XeOF_4 .

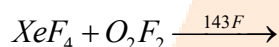
1+1+1+1=4

OR

(a) Write general electronic configuration of elements of group-16.

(b) Why the value of electron gain enthalpy of chlorine is higher than fluorine? Explain.

(c) Complete the following reaction:



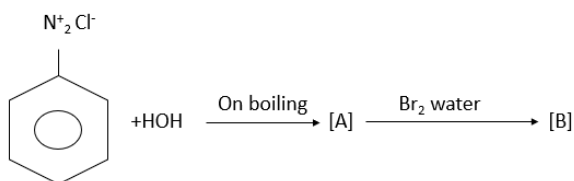
(d) Draw the structure of H_3PO_3 .

30. (a) Write structural formula of isobutyl alcohol.

(b) What happens when diethyl ether reacts with chlorine in presence of sunlight? Give chemical reaction.

(c) Complete the following reaction sequence and write the names of products [A] and [M].

1+1+2=4



OR

- Write structural formula of tert-butyl alcohol.
- What happens when the vapour of diethyl ether is passed over alumina at 653 K temperature? Give chemical reaction.
- complete the following reaction sequence and write the names of products [A] and [B].

